513 Project 1 Deliverables

Emmett Williamson

November 7, 2026

Github Repository

`https://github.com/emmwill-star/EE513-project-01`

Resume Line

Developed and tested a calibration pipeline for camera systems. Pipeline determines intrinsic camera geometry, MTF50, noise, dynamic range, and color correction.

Decisions and Verification

LLM Usage

I used a mixture of Agentic and Code focused LLM tools including Snowflake Cortex Code and Github Copilot. The tools were used to trace calculations through the notebook, create helper functions, draft the report check uncertainty definitions and compare results with manufacturer information. AI assisted checks included independent recalculation of selected summaries and LaTeX compilation. All automated tests were created by the AI tools. I worked by questioning the results and asking for further information and explanation when I did not understand the results.

Deliverable 1

I chose subpixel checkerboard calibration with radial and tangential distortion, keeping images in consistent sensor coordinates and fixing the third radial coefficient. I rejected treating fit errors as a complete uncertainty budget or comparing physical focal length directly with Apple's equivalent focal length. Existing tests check zero residuals for a straight grid, correction of synthetic distortion, image dimension ordering and reprojection error against known offsets. The notebook also compares original and corrected checkerboard straightness.

Deliverable 2

I used black subtracted RAW green samples and slanted edge MTF, avoiding demosaicing and gamma processing. I reported frequencies per original RAW pixel and selected the worst corner by its average across captures, not by averaging individual minima. Existing tests cover synthetic target selection, uneven illumination, missing targets, batch completeness, frequency scaling and worst corner aggregation. Recomputed summaries agreed with the saved results. Review established that pooled corner spread mixes spatial variation with repeatability.

Deliverable 3

I chose paired frame variance to suppress common fixed pattern and used dark frame differences rather than the single frame spatial variance intercept for read noise. Gain and dark current came from linear fits. Checks included comparison with single frame results and conversion of the DNG vendor noise model into matching units. Review found that read noise uncertainty includes only gain fit error, dark pairs share frames, and unequal exposures can introduce additional noise. Dark signals were also close to the black offset.

Deliverable 4

I defined range using metadata saturation above black divided by the dark noise floor, with an explicit read only noise threshold. I rejected the saved zero uncertainty because it resulted from taking the standard deviation of one scalar. The report therefore labels the result provisional.

Deliverable 5

I chose per channel flat field correction, white balancing, capture averaging and a least squares color matrix evaluated against the specified ColorChecker reference under D50. Existing tests check masked patch statistics against known values. Review found that the maximum error uncertainty incorrectly reused across patch spread; that error bar was rejected. Because the same patches determine and evaluate the matrix, the improvement is in sample rather than independent validation.

Table 1: Sensor Properties and Capture Specifications.

Parameter	Value	Unit
`Lens ID`	iPhone 16 back camera 5.96mm f/1.6	
`App`	RawCam	
`CFA Pattern`	RGGB	
`Raw Image Width`	4032	pixels
`Raw Image Height`	3024	pixels

Deliverable 1: Geometry Intrinsics and Lens Distortion

Method. 19 DNG captures are captured using the same rear camera and a 9×6 internal-corner checkerboard. Corners were refined to subpixel precision on demosaiced, unrotated images; `calibrateCameraExtended` determines the camera intrinsic parameters from the detected corners. The distortion model uses radial terms $1 + k_1 r^2 + k_2 r^4$ in normalized coordinates and tangential coefficients p_1, p_2; $k_3 = 0$ was fixed, and assumed to be negligible. Checkerboard spacing was measured in mm using a ruler. A speculative absolute uncertainty of 0.25 mm was used based on the precision of the ruler. Image dimensions and capture counts are exact metadata/counts, not uncertain measurements.

Table 2: Geometry results. Uncertainties are fit standard deviations, except the RMS row, which reports across-view scatter.

Parameter	Value	Uncertainty	Unit
Mean focal length $\bar{f}$	3125.3	4.9	px
Physical focal length (conditional)	6.25	0.01	mm
Principal point c_x	1990.7	7.1	px
Principal point c_y	1524.7	8.3	px
Radial k_1	0.106	0.005	dimensionless
Radial k_2	−0.192	0.016	dimensionless
Tangential p_1	0.0035	0.0011	dimensionless
Tangential p_2	−0.0068	0.0010	dimensionless
Pooled reprojection RMS	2.11	0.61	px
Checkerboard square size	20.00	0.25	mm

Calculations and uncertainty. Intrinsic uncertainties are OpenCV's approximate 1σ fit errors, not repeat-calibration standard deviations or full systematic bounds. The notebook averages the two focal lengths and propagates their errors:

$$\bar{f} = \frac{f_x + f_y}{2}, \qquad u(\bar{f}) = \frac{\sqrt{u(f_x)^2 + u(f_y)^2}}{2}, \qquad f_{\mathrm{mm}} = p\bar{f}.$$

The first propagation neglects axis covariance. The assumed $p = 0.002\,\mathrm{mm/px}$ has no verified tolerance; hence the physical focal-length uncertainty is *fit-only*, conditional on that scale. If pitch uncertainty were known, independent propagation would give $u(f_{\mathrm{mm}})^2 = p^2 u(\bar{f})^2 + \bar{f}^2 u(p)^2$. Reprojection RMS pools squared Euclidean corner residuals over all views; its accompanying spread is `np.std(errorValues)` (population standard deviation of per-view RMS), not a confidence interval on the pooled RMS. The notebook's before/after straightness check is in-sample, not independent validation.

Manufacturer comparison. Apple's iPhone 16 specification lists a nominal 26 mm equivalent main lens, not its physical focal length, and gives no tolerance. Matching image diagonals to the 36×24 mm reference format gives $f_{\mathrm{eq}} = \bar{f}\sqrt{36^2 + 24^2}/\sqrt{4032^2 + 3024^2} = (26.83 \pm 0.04)\,\mathrm{mm}$ (fit-only 1σ), or $(3.19 \pm 0.16)\%$ above that nominal value. RAW versus advertised field-of-view/crop conventions, focus changes and calibration-model bias are possible contributors, but were not isolated or further tested.

Deliverable 2: Spatial Resolution MTF

Method. Five DNG captures supplied black-subtracted samples from one Bayer green plane, without demosaicing or gamma encoding. Near-vertical target edges were selected at the center and four corners. Samples were aligned by perpendicular distance to a fitted edge and binned into a four-times-oversampled edge-spread function (ESF). Differentiation gives the line-spread function (LSF); its Hann-windowed Fourier magnitude is normalized at zero frequency:

$$\mathrm{MTF}(f) = \frac{|\mathcal{F}\{w\,\mathrm{LSF}\}(f)|}{|\mathcal{F}\{w\,\mathrm{LSF}\}(0)|}, \qquad \mathrm{MTF}(f_{50}) = 0.5.$$

The first downward half-response crossing is linearly interpolated. Green-plane spacing is twice the original RAW pixel pitch, so its axial Nyquist limit is 0.25 cycles/original pixel. The pitch assumed for intermediate physical units cancels when converting back to cycles/original pixel.

Table 3: MTF50 in cycles/original RAW pixel; uncertainties are population standard deviations, with pooling defined below.

Parameter	Value	Uncertainty	Unit
`Center MTF50`	0.228	0.008	cycles/pixel
`Corner Mean MTF50`	0.191	0.017	cycles/pixel
`Worst (Top-right) MTF50`	0.170	0.007	cycles/pixel

Uncertainty and interpretation. The reported uncertainties are population standard deviations. Center and worst-corner spreads each use five captures; the corner mean pools twenty values, mixing spatial variation with capture repeatability. The top-right corner has the lowest five-capture mean. Corners resolve less detail than the center in the tested edge direction. These are target-plus-camera green-channel responses, not isolated lens MTF; printed-edge quality, alignment, aliasing and estimator bias are not bounded by the reported scatter.

Manufacturer comparison. Apple's published specification provides no comparable MTF50 figure. Its nominal megapixel count describes sampling, not contrast transfer, so it cannot establish agreement with these RAW edge measurements.

Deliverable 3: Noise

Method. At reference-capture ISO 40, central 128×128 green-sample patches from eleven flat-field pairs provided black-subtracted mean μ and temporal variance $v = \mathrm{Var}(A - B)/2$. Five levels within the selected linear range were fitted to $v = g\mu + b$; g is conversion gain in DN/electron. Differencing suppresses common fixed pattern. Five dark captures supplied ten overlapping pairs, giving $\sigma_{\mathrm{DN}} = \sqrt{\langle v_{\mathrm{dark}} \rangle}$ and $\sigma_e = \sigma_{\mathrm{DN}}/g$. Dark current is the slope of mean black-subtracted dark signal in electrons versus exposure time.

Table 4: Noise results; uncertainty components are defined below.

Parameter	Value	Uncertainty	Unit
`Conversion Gain`	0.262	0.006	DN/e^-
`Read Noise (RMS)`	6.863	0.160	e^-
`Dark Current`	2.899	0.537	e^-/pixel/s
`Black Level`	528.000	0.000	DN
White Level	4095		DN
ISO	40		
Vendor Noise Found	True		

Uncertainty and comparison. Gain and dark-current uncertainties are 1σ slope standard errors from fit covariance; dark-current error holds gain fixed. Read-noise error is only $u(\sigma_e) = \sigma_e u(g)/g$, omitting uncertainty in the dark variance. Shared frames make the ten pairs non-independent; unequal exposures also admit dark-current shot noise and exposure-dependent pattern. Dark means lie close to the black offset, so the current slope is provisional and black-level drift is not bounded.

The DNG vendor `NoiseProfile`, interpreted as normalized variance $ax+b_v$, predicts $g_v = aS \simeq 0.344$ DN/e^- and $\sigma_{e,v} = \sqrt{b_v S^2}/g_v \simeq 10.26\, e^-$, where S is the metadata white-minus-black range. These are nominal model predictions without supplied tolerances, both higher than the fitted results. The discrepancy may reflect central-patch selection, the restricted signal range or dark-frame conditions, but vendor characterization conditions are unspecified and the cause cannot be isolated.

Deliverable 4: Dynamic Range

Calculation and uncertainty. Using metadata white level $W = 4095$ DN and black level $B = 528$ DN, the notebook defines $S = W - B$ and $D = \log_2(S/\sigma_{\mathrm{DN}})$. These code levels are metadata, not independently measured saturation or black-offset calibrations. The low end is one RMS dark-noise level: SNR unity in a read-noise-only convention, not exactly unity when shot noise is included.

Table 5: Provisional dynamic-range estimate; uncertainty is not established.

Parameter	Estimate	Uncertainty	Unit
Dynamic range	10.95	Not estimated	stops

The JSON zero uncertainty comes from applying `std` to a single scalar; it is not evidence of zero measurement error. A defensible interval requires repeated noise-floor and saturation/offset characterization. Apple's cited specification gives no comparable single-frame RAW stop count; its HDR features do not validate this provisional estimate.

Deliverable 5: Color Accuracy

Method. Three chart captures were black-subtracted, normalized and per-channel flat-field corrected, then their linear RGB patch means were averaged. White patch 19 supplied white-balance gains. The baseline treats balanced camera RGB as linear sRGB with Bradford adaptation to D50; it is not Apple's rendered color pipeline. A least-squares 3×3 matrix maps balanced camera RGB to reference XYZ. Both results are converted to Lab under D50 and evaluated using CIEDE2000 (ΔE_{00}), against *ColorChecker24 – After November 2014*.

Table 6: Dimensionless color errors. Spread is across chart patches, not measurement uncertainty; the maximum has no estimated uncertainty.

Parameter	**Value**	**Uncertainty**
Mean before matrix	7.63	3.69
Mean after matrix	2.14	0.88
Worst after (orange, patch 7)	3.58	0.88

Uncertainty and interpretation. The listed spreads are population standard deviations across all 24 patches, not confidence intervals on the mean or repeat-capture errors. The JSON's worst-patch error bar repeats the across-patch spread over all patches, and has the same uncertainty. All 24 patches were used for both fitting and evaluation: the reduced mean error demonstrates in-sample correction, not independent color accuracy. Illumination, chart-reference and flat-field errors remain unquantified. Apple's cited specification gives no comparable CIEDE2000 tolerance; wide-color support is not an accuracy guarantee.